

CIRCUITS

VOLTAGE
(VOLT V) $\frac{\text{Joules}}{\text{Coulomb}}$

$$V = \frac{dw}{dq}$$

RESISTANCE
(OHMS Ω) $\frac{\text{Volts}}{\text{AMPS}}$

$$R = \frac{V}{I}$$

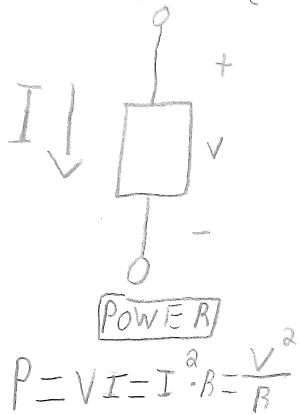
$$V = IR$$

CURRENT
(AMPS I) $\frac{\text{Coulombs}}{\text{second}}$

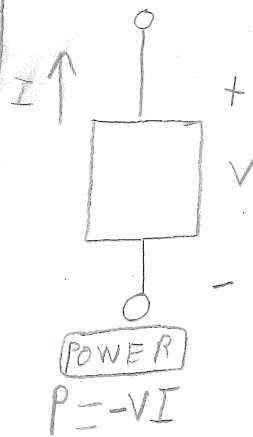
$$i = \frac{dq}{dt}$$

PASSIVE SIGN CONVENTION

PASSIVE SOURCE
(~)



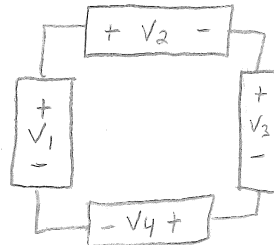
ACTIVE SOURCE (⊕⊕)



KIRCHHOFFS LAWS

VOLTAGE

$$\sum_{\text{Loop}} V = 0$$



$$\text{KVL: } -V_1 + V_2 + V_3 + V_4 = 0$$

CURRENT

$$\sum I_{in} = \sum I_{out}$$



$$I_2 + I_3 = I_1 + I_4$$

CAPACITANCE
(FARADS)(F) $\frac{\text{COULOMBS}}{\text{VOLT}}$

$$C = \frac{q}{V}$$

CONDUCTANCE
(SIEMENS) $\frac{1}{\text{OHMS}}$

$$G = \frac{1}{R}$$

POWER

NEGATIVE POSITIVE
SUPPLIED **ABSORBED**
 $P = -IV$ $P = IV$

INDUCTANCE
(HENRIES H)

$$L = \frac{\lambda}{I} = \frac{N\phi}{I}$$

λ = FLUX LINKAGE

N = # OF TERMS

ϕ = TOTAL FLUX

SERIES

$$Q_{EQ} = Q_1 = Q_2$$

$$\Delta V = \Delta V_1 + \Delta V_2$$

$$R_{EQ} = R_1 + R_2 + R_3$$

$$I = I_1 = I_2$$

$$\frac{1}{C_{EQ}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$L_{EQ} = L_1 + L_2 + L_3$$

PARALLEL

$$Q = Q_1 + Q_2$$

$$\Delta V = \Delta V_1 = \Delta V_2$$

$$\frac{1}{R_{EQ}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$I = I_1 + I_2$$

$$C_{EQ} = C_1 + C_2 + C_3$$

$$\frac{1}{L_{EQ}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$$

COMPONENTS

VOLT



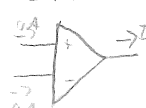
RESISTOR



INDUCTORS



OPP AMP



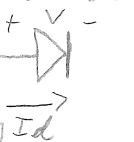
CURRENT



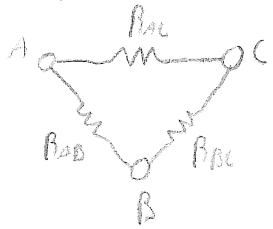
CAPACITOR



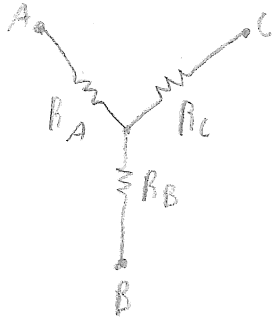
DIODE



DELTA (Δ)



WYE (Y) NETWORK



$$R_A = \frac{R_{AB} \cdot R_{AC}}{R_{AB} + R_{AC} + R_{BC}}$$

$$R_B = \frac{R_{AB} \cdot R_{BC}}{R_{AB} + R_{AC} + R_{BC}}$$

$$R_C = \frac{R_{AC} \cdot R_{BC}}{R_{AB} + R_{AC} + R_{BC}}$$

$$R_{AB} = \frac{R_{AB0} + R_{AB1} + R_{AB2}}{R_C}$$

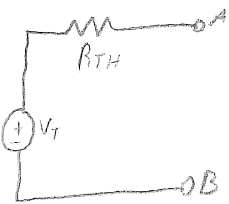
$$R_{BC} = \frac{R_{AB0} + R_{AB1} + R_{AB2}}{R_A}$$

$$R_{AC} = \frac{R_{AB0} + R_{AB1} + R_{AB2}}{R_B}$$

$$R_{\Delta} = 3R_{Wye} \text{ IF SAME } \Omega$$

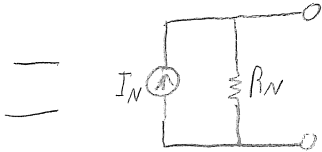
ANY LINEAR CIRCUIT CAN BE REPRESENTED

THEVENIN



$$V_T = I_N \cdot R$$

NORTON



$$R_{TH} = R_N$$

PHASORS

$$e^{jx} = \cos(x) + j\sin(x)$$

$$\cos(x) = \Re\{e^{jx}\}$$

$$v(t) = V_m \cos(\omega t + \phi) = \tilde{v} = V_m \angle \phi = V_m e^{j\phi}$$

$$\text{OHMS LAW } \tilde{v} = Z \tilde{i}$$

$Z = \text{IMPEDANCE}$

$$\tilde{V}_L = L j\omega \tilde{I}_L$$

$$Z_L = \frac{\tilde{V}_L}{\tilde{I}_L} = j\omega L$$

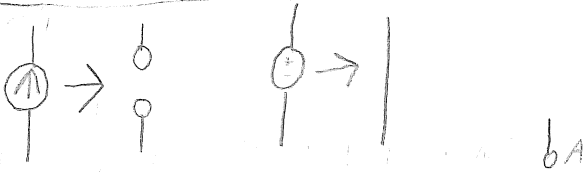
$$\tilde{I}_C = C j\omega \tilde{V}_C$$

$$Z_C = \frac{\tilde{V}_C}{\tilde{I}_C} = \frac{-j}{\omega C}$$

$$p(t) = v(t) i(t)$$

NO DEPENDENT SOURCE

① REMOVE INDEPENDENT SOURCES



② SOLVE FOR R_{eq} BETWEEN a & b

$$Z = R + jX \Omega$$

$R = \text{RESISTANCE}$

$$X > 0 \rightarrow L$$

$X = \text{REACTANCE}$

$$X < 0 \rightarrow C$$

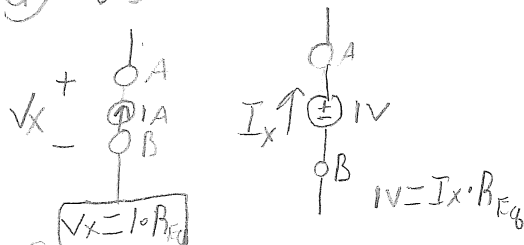
$Z = \text{IMPEDANCE}$

$$X = 0 \rightarrow R$$

WITH DEPENDENT SOURCE

① REMOVE INDEPENDENT SOURCES

② USE TEST SOURCE



$$V_x = I_x \cdot R_{eq}$$

③ SOLVE FOR V_x OR I_x AND SOLVE FOR R_{eq}

① CONVERT TIME DOMAIN TO FREQUENCY

② USE COMPLEX ALGEBRA TO FIND QUANTITY OF INTEREST

③ FIND PHASOR SOLUTION $\tilde{v}$ or $\tilde{i}$

④ CONVERT BACK TO TIME DOMAIN ω HAS TO BE GIVEN $\cos(\omega t + \text{---})$

★ IF INDEPENDENT SOURCE USE SUPERPOSITION
WHEN SOLVING INDEPENDENT SOURCE USE DC-STEADY STATE
★ R_{eq} IS VERY HELPFUL

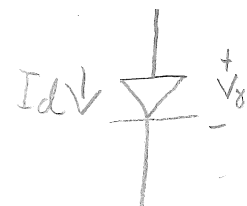
SOLVING DIODES

① ASSUME STATE

ON

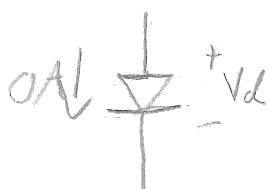
$$V_d = 5V$$

$$I_d > 0$$



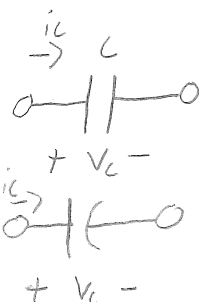
OFF

$$V_d < V_g$$



② IF CONDITIONS ARE FALSE TRY OTHER STATE

GIVEN



$$i_c = C \frac{dV_c}{dt}$$

SOLVING CAPACITORS

$$t = 0^-$$

$$i_c = 0A$$

① SOLVE V_c

$$t = \infty$$

$$i_c = 0A$$

③ SOLVE V_c

OPEN

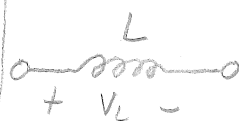
$$t = 0^+$$

$$V_c(0^-) = V_c(0^+)$$

② SOLVE i_c

$$t > 0$$

$$i_c = C \frac{dV_c}{dt}$$



$$V_L = L \frac{di_L}{dt}$$

SOLVING INDUCTORS

$$t = 0^-$$

SHORT

$$V_L = 0V$$

① SOLVE i_L

$$t = \infty$$

SHORT

$$V_L = 0V$$

③ SOLVE i_L

$$t = 0^+$$

$$i_L(0^-) = i_L(0^+)$$

② SOLVE V_L

$$t > 0$$

$$V_L = L \frac{di_L}{dt}$$

SECOND ORDER ODE

$$\frac{d^2 y}{dt^2} + A \frac{dy}{dt} + By = K$$

1) FIND INITIAL CONDITION $y(0^-)$, $y'(0^-)$

• $t=0^- \rightarrow$ DC SS $\rightarrow$ C OPEN, L SHORT

• FIND WHAT CANNOT CHANGE INSTANTANEOUSLY FROM $t=0^-$ TO $t=0^+$

• $t=0^+ \rightarrow$ WE KNOW $V_C(0^+)$ OR $i_L(0^+)$ AND THAT $V_C' = \frac{i_C(0^+)}{C}$ & $i_L' = \frac{V_L(0^+)}{L}$

• SOLVE FOR i_C AND/OR V_L

2) FIND STEADY-STATE: $y(\infty)$

• $t=\infty \rightarrow$ DC SS $\rightarrow$ C OPEN, L SHORT

• FIND $V_C(\infty)$ OR $i_L(\infty)$

3) FIND DIFFERENTIAL EQUATION RELATING the output to input, 2 EQUATIONS 2 UNKNOWN

4) WRITE CHARACTERISTICS EQUATION

a) s_1 & s_2 ARE REAL AND DISTINCT (OVERDAMPED)

$$f(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + f(\infty)$$

b) s_1 & s_2 ARE REAL AND EQUAL (CRITICALLY DAMPED)

$$f(t) = A_1 t e^{s_1 t} + A_2 e^{s_1 t} + f(\infty)$$

c) s_1 & s_2 ARE COMPLEX CONJUGATES, $\sigma \pm j\omega_d$ (UNDERDAMPED)

$$f(t) = A_1 e^{-\sigma t} \cos(\omega_d t) + A_2 e^{-\sigma t} \sin(\omega_d t) + f(\infty)$$

FIRST ORDER

① ANALYZE $t=0^-$ CIRCUIT (CAPACITOR OPEN, INDUCTOR SHORT)

FIND $V_C(0^-)$ OR $i_L(0^-)$

② ANALYZE $t=0^+$ USING $V_C(0^-) = V_C(0^+)$ OR $i_L(0^-) = i_L(0^+)$

FIND $f(0^+)$

③ ANALYZE $t=\infty$ CIRCUIT (CAPACITOR OPEN, INDUCTOR SHORT)

FIND $f(\infty)$

④ ANALYZE $t>0$. REMOVE L OR C. (MARK TERMINALS A AND B)

- TURN OFF INDEPENDENT SOURCES

- FIND $R_T = R_{\text{EQUIV}}$ BETWEEN A AND B

$$\tau = R_T \cdot C \quad \tau = \frac{L}{R_T}$$

⑤ WRITE FINAL EQUATION

$$f(t) = [f(0^+) - f(\infty)] e^{-\frac{t}{\tau}} + f(\infty)$$

DAMPING CONDITION FOR RLC

$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$$

• OVER DAMPED: $\zeta > 1$; $R^2 > \frac{4L}{C}$

• CRITICALLY DAMPED

$$\zeta = 1; R^2 = \frac{4L}{C}$$

• UNDER DAMPED

$$\zeta < 1; R^2 < \frac{4L}{C}$$

AC

POWER

$$P(t) = V(t)i(t)$$

REAL POWER

$$P = \frac{1}{T} \int_0^T P(t) dt$$

AVERAGE POWER

$$P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i)$$

$$P = V_{RMS} I_{RMS} \cos(\theta_v - \theta_i)$$

APPARENT POWER

$$V_{RMS} I_{RMS}$$

POWER FACTOR

$$\cos(\theta_v - \theta_i)$$

$$P = V_{RMS} I_{RMS} \cos(\theta_v - \theta_i)$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

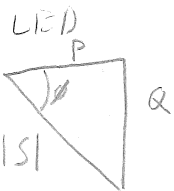
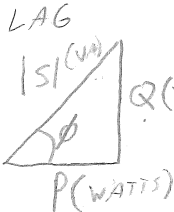
POWER TRIANGLE

LAGGING (CURRENT LAGS VOLTAGE)

$\theta > 0, Q > 0, \theta_v > \theta_i, \text{INDUCTIVE}$

LEADING (CURRENT LEADS VOLTAGE)

$\theta < 0, Q < 0, \theta_v < \theta_i, \text{CAPACITIVE}$



$$\cos(\theta) = \frac{P}{|S|}$$

$$PF(\text{POWER FACTOR}) = \frac{P}{|S|}$$

$$\theta = \theta_v - \theta_i = \cos^{-1}(PF)$$

$$PF = \cos(\theta_v - \theta_i) = \cos(\theta) = \cos(\theta_z)$$

INSTANTANEOUS POWER

WHERE V & I ARE COSINES

$$P(t) = \frac{V_m I_m}{2} [\cos(\omega t + \theta_v + \theta_i) + \cos(\theta_v - \theta_i)]$$

RMS

- SQUARE
- MEAN
- √

$$V_{RMS} = \frac{V_m}{\sqrt{2}} \text{ WHEN V \& I ARE COSINES}$$

$$I_{RMS} = \frac{I_m}{\sqrt{2}}$$

(RMS) COMPLEX POWER

$$S = \tilde{V}_{RMS} \tilde{I}_{RMS}^* = V_{RMS} I_{RMS} e^{j(\theta_v - \theta_i)}$$

$$S = P + jQ$$

P IS REAL POWER

Q IS REACTIVE POWER

ALL CALCULATIONS NEED RMS

$$S = \tilde{V} \tilde{I}^* = Z \tilde{I} \tilde{I}^* = |\tilde{I}|^2 Z$$

$$S = \tilde{V} \tilde{I}^* = \tilde{V} \left[\frac{\tilde{V}}{Z} \right]^* = \frac{|\tilde{V}|^2}{Z^*} = \frac{|\tilde{V}|^2}{Z} = |S| \angle \phi_s = R + jQ$$

INDUCTOR

$$\tilde{V} = j\omega L \tilde{I} \rightarrow \theta_z = 90^\circ$$

$$Z_Y = \frac{Z_\Delta}{3}$$

WYE (Y)

$$V_{Line} = \sqrt{3} V_{Phase} \angle 30^\circ$$

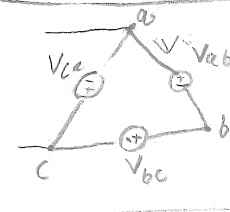
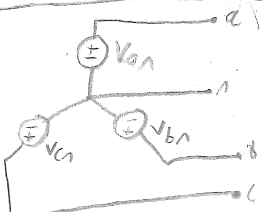
$$I_{Line} = I_{Phase}$$

DELTA (Δ)

$$V_{Line} = V_{Phase}$$

$$I_{Line} = I_{Phase} \angle -30^\circ$$

WHEN SOLVING BEPACCE Δ WITH Y AND CONNECT NATURAL



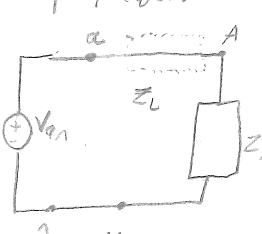
$$\tilde{V}_{ab} = \tilde{V}_{an} [\sqrt{3} \angle 30^\circ]$$

V_line is a & b

V_phase is a & n

$$\tilde{I}_{ga} = \sqrt{3} \tilde{I}_{ab} \angle -30^\circ$$

Y-Y equiv



$$S_{1\phi} = \tilde{V}_{an} \tilde{I}_a^* = \frac{|\tilde{V}_{an}|^2}{Z_L} = |\tilde{I}_a|^2 Z_L$$

$$S_{1\phi} = |\tilde{I}_a|^2 Z_Y \text{ or } S_{1\phi} = \frac{|\tilde{V}_{ab}|^2}{Z_\Delta}$$

FIND 3 Phases

$$S_{3\phi} = 3 S_{1\phi} = P_{3\phi} + jQ_{3\phi}$$

MAX POWER TRANSFER

$$Z_T = R_T + jX_T$$

$$\tilde{I}_T = \frac{\tilde{V}_T}{Z_T + Z_L} = \frac{\tilde{V}_T}{(R_T + R_L) + (jX_T + jX_L)}$$

$$S_L = |\tilde{I}|^2 Z_L$$

$$|\tilde{I}|^2 = \frac{|\tilde{V}_T|^2}{(R_T + R_L)^2 + (X_T + X_L)^2}$$

$$S_L = \frac{|\tilde{V}_T|^2}{(R_T + R_L)^2 + (X_T + X_L)^2} (R_L + jX_L)$$

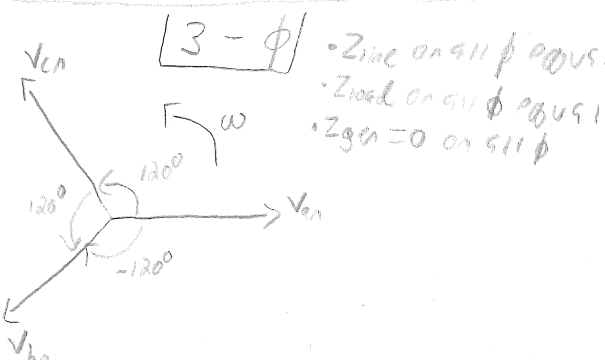
REAL POWER TO LOAD

$$P_L = \frac{|\tilde{V}_T|^2 R_L}{(R_T + R_L)^2 + (X_T + X_L)^2}$$

P_L IS MAX WHEN

$$X_L = -X_T \text{ and } R_L = \sqrt{R_T^2 + (X_T + X_L)^2}$$

P_L IS MAX WHEN $Z_L = R_T - jX_T = Z_T^*$



SELF INDUCTANCE

SINGLE COIL

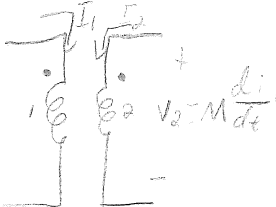
$$V = N \frac{d\phi}{dt}$$

$$V = N \frac{d\phi}{dt} \cdot \frac{dI}{dt}$$

$$L = N \frac{d\phi}{dI} \rightarrow$$

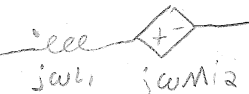
$$\rightarrow V = L \cdot \frac{dI}{dt}$$

SELF INDUCTANCE



THE VOLTAGE ACROSS IS $V = L \frac{dI}{dt} + V = M \frac{dI}{dt}$

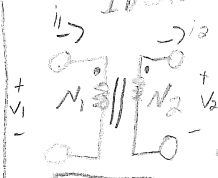
TO CONVERT TO PHASOR



- 1) ADD DEPENDENT SOURCE
- 2) CONVERT TO PHASOR
- 3) DEFINE $\tilde{I}_1$ & $\tilde{I}_2$
- 4) Put positive turns on a side of coil on dependent source
- 5) Find value of each depend source
- 6) SOLVE AS USUAL
- 7) CONVERT BACK

TRANSFORMERS

IDEAL COUPLING



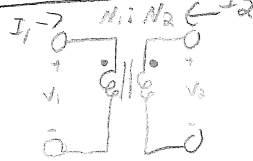
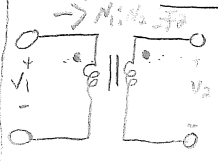
$$V_2 = \pm N_2 \frac{d\phi}{dt}$$

$$I_1 = \pm N_1 \frac{d\phi}{dt}$$

$$\frac{V_2}{V_1} = \pm \frac{N_2}{N_1} = \pm \frac{I_1}{I_2}$$

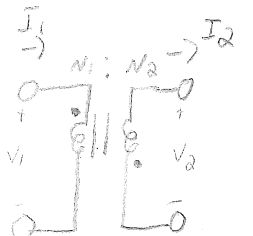
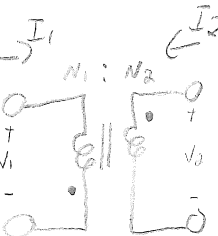
N: TURNS RATIO

SIGN CONVENTION



$$\frac{V_2}{V_1} = \pm \frac{N_2}{N_1} = \pm \frac{I_1}{I_2}$$

$$\frac{V_2}{V_1} = \pm \frac{N_2}{N_1} = \pm \frac{I_1}{I_2}$$



$$\frac{V_2}{V_1} = \pm \frac{N_2}{N_1} = \pm \frac{I_1}{I_2}$$

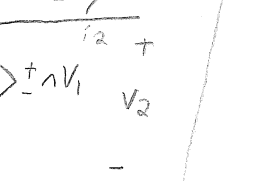
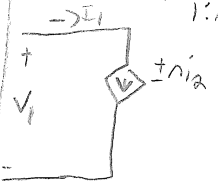
$$\frac{V_2}{V_1} = \pm \frac{N_2}{N_1} = \pm \frac{I_1}{I_2}$$

DOES ON SAME SIDE V-POSITIVE CURRENTS GOING IN AND OUT OF DOT I=POSITIVE

FOR TOP LEFT



EQUALS



FREQUENCY RESPONSE

TIME

FREQUENCY

R	$V(t) = i(t)R$	$\tilde{V} = \tilde{I}R$
L	$V(t) = L \frac{di}{dt}$	$\tilde{V} = j\omega L \tilde{I}$
C	$i(t) = C \frac{dV}{dt}$	$\tilde{V} = \frac{1}{j\omega C} \tilde{I}$

THE LAPLACE TRANSFORM

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

$$\mathcal{L}\{f(t)\} = sF(s) - f(0^+)$$

S DOMAIN

$$\frac{R}{V(s) = I(s)R}$$

$$\frac{L}{V(s) = L(sI(s) - i(0^+))}$$

$$Z_L(s) = sL$$

$$\frac{C}{I(s) = C(sV(s) - v(0^+))}$$

$$Z_C(s) = \frac{1}{sC}$$

FREQUENCY RESPONSE

$$H(s) = \frac{\text{OUTPUT}(s)}{\text{INPUT}(s)}$$

$$s = j\omega, G_{dB} = 20 \log_{10} \left(\frac{P_2}{P_1} \right)$$

$$H(j\omega) = |H(j\omega)| \angle H(j\omega)$$

$$G_{dB} = 20 \log_{10}(A_V)$$

$$G_{dB} = 20 \log_{10}(|H(j\omega)|)$$

$$\text{PLOT } H(j\omega) \text{ vs } \omega$$

POSSIBLE TERMS FROM FACTORING

- 1) CONSTANT: K $G_{dB} = 20 \log_{10}(K)$
- 2) POLE/ZERO AT THE ORIGIN: s^N
 - Denominator ($N < 0$) $\rightarrow$ POLE
 - Numerator ($N > 0$) $\rightarrow$ ZERO
 - UNSTABLE
- 3) REAL POLES OR ZEROS $\left(\frac{s}{\omega_0} + 1 \right)^N$
 - $|H(j\omega)| = \left(1 + \frac{\omega^2}{\omega_0^2} \right)^{N/2}$
- 4) COMPLEX CONJUGATE POLES OR ZEROS $\left(\frac{s^2}{\omega_0^2} + 2\zeta \frac{s}{\omega_0} + 1 \right)^N$
 - $\omega_0 = \text{BREAK F}$
 - $\zeta = \text{DAMPING RATIO}$

$$\angle H(j\omega) \text{ in degrees}$$

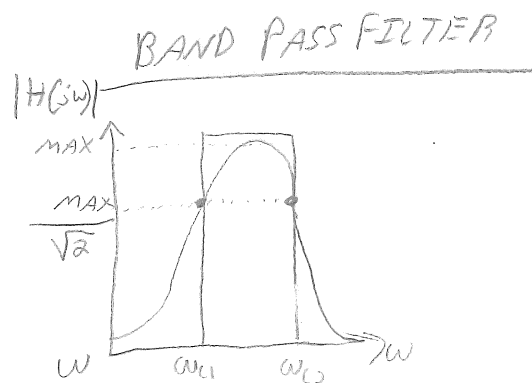
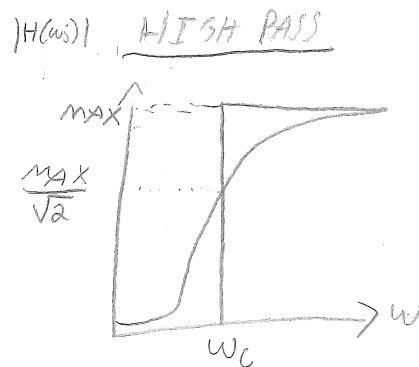
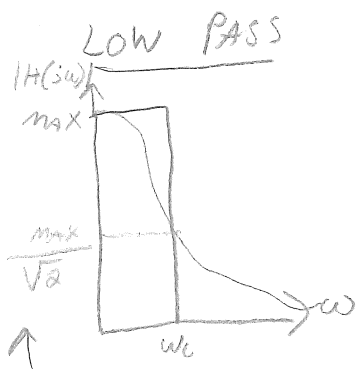
$$n = \frac{\text{SLOPE (dB/dec)}}{20}$$

$$|H(j\omega)|^2 = \omega^n$$

TO SOLVE FOR ω

$$|H(j\omega)|^2 = \left(\frac{\text{max}}{\sqrt{2}} \right)^2$$

FILTERS



1/2 PF

(CHARACTERIZING FILTERS)

1) TYPE: LP, HP, BP

2) ORDER: # of ENFOLDY STAGE

3) PASS BAND GAIN: MAX GAIN

4) CUTOFF FREQUENCY: w_c

SECOND ORDER FILTER

$N=0 \Rightarrow H(s) = \frac{Ks^0}{s^2 + 2\zeta w_0 s + w_0^2}$

LPF

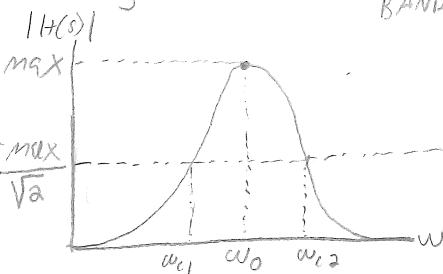
DC GAIN = $\frac{K}{w_0^2}$

∞ f GAIN = $\emptyset$

$N=1 \Rightarrow H(s) = \frac{Ks^1}{s^2 + 2\zeta w_0 s + w_0^2}$

BPF, 2nd order

BANDWIDTH



$|H|_{max} = \frac{K}{2\zeta w_0}$

$|H(jw)|_{max} = \text{derivative} = 0$

POLE $w = w_0$

TO FIND CUTOFF w_c

$|H(jw)| = \frac{max}{\sqrt{2}}$

FIRST ORDER FILTER

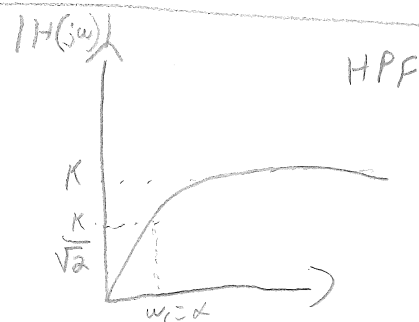
1) $H(s) = \frac{K}{s+a} = \frac{SKw}{jw+a}$

DC GAIN: $|H(j0)| = \emptyset$

∞ f GAIN: $|H(jw)| = K$

$|H(jw)| = \left| \frac{SKw}{jw+a} \right| = \left(\frac{K}{\sqrt{2}} \right)^2$

$w_c = a$



2) $H(s) = \frac{K}{s+a}$

DC GAIN: $|H(j0)| = K/a$

∞ f GAIN: $|H(jw)| = \emptyset$

$w_c = a$

